Count Permutation

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	512 megabytes

Alice loves doing research on permutations. For a n-permutation $P = (p_1, p_2, ..., p_n)$, she defines $w(P) = |S|$, where $S = \left\{p_i - p_{(i \bmod n)+1} | 1 \le i \le n\right\}$. For example, if $P = (2, 4, 5, 3, 1)$, then $S = \{2, -1, -2\}, w(P) = |S| = 3$.

Alice hopes S to be as simple as possible, so she wants to find all n-permutation P such that $w(P)$ is minimal (She defines these permutations as "good permutation"). However, the number of good permutations may be too large. As a result, she writes down three numbers m, s, t and wants you to find out the number of good permutations $P = (p_1, p_2, ..., p_n)$ that $p_1 = s$ and $p_m = t$.

Input

One line that contains 4 integers $n, m, s, t, (3 \le n \le 10^{12}, 2 \le m \le n, 1 \le s, t \le n, s \ne t)$.

Output

One line, an integer, denoting the answer.

Examples

standard input	standard output
3 2 1 3	1
4 3 1 3	2

Note

A n-permutation is an integer sequence whose length is n and each element of $\{1, 2, ..., n\}$ appears exactly once.